

Piyush Dusane

Passionate data engineer with 2.5+ years of experience in AWS serverless technologies, including Lambda, Glue, DynamoDB, and API Gateway. Proficient in Python and Java for building scalable APIs and automating ETL pipelines. Expertise in designing event-driven architectures, enabling seamless data integration, and delivering reliable, customer-focused solutions. Skilled in technical analysis and solution architecture to drive efficiency and innovation.

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Work Experience

Yash Technology

Software Engineer

Jan 2025 - Present

- Designed and architected an end-to-end Medallion Data Lake framework Raw landing zone, Exploratory transformation layer, and Curated business layer using Amazon S3, Apache Iceberg, Amazon S3 Tables, and AWS Glue with ACID transaction support and full schema evolution.
- Worked with Apache Iceberg and Amazon S3 Tables as the primary open table format enabling time-travel queries, snapshot isolation, partition evolution, and incremental reads for downstream consumers.
- Implemented AWS Distro for OpenTelemetry (ADOT) for end-to-end pipeline observability metrics, traces, and logs exported to Amazon CloudWatch and X-Ray for proactive monitoring.
- Contributed to SageMaker Unified Studio (SMUS) integration for unified data and ML workflows and participated in DevOps agent automation for zero-touch infrastructure provisioning.
- Drove AI enablement using AWS Kiro AI-assisted pipeline design, infrastructure code generation, and intelligent workflow automation
- Integrated Amazon Macie for automated PII/sensitive data discovery, and AWS Clean Rooms for secure cross-team and cross-account analytics without raw data exposure.

Verolt Technology

Trainee Engineer

April 2024 - Nov 2024

- Designed and implemented a cost monitoring architecture to fetch AWS Glue job execution costs across multiple clients by integrating AWS Glue and CloudWatch APIs, enabling centralized cost visibility and optimization on QuickSight
- Developed pagination strategy to limit API responses to the first 100 records, significantly enhancing response times and user experience.
- Implemented advanced unit and integration testing strategies, increasing system reliability by resolving edge-case scenarios

Intern

Feb 2024 - April 2024

- Wrote Python scripts for AWS Glue jobs, focusing on data manipulation and ETL transformations.
- End to end deployment of all the functionalities using GitHub

Education

SITS - Pune

BE - Information Technology - 7.8 GPA

2023

Skills

- Programming Languages: Python, Java, Embedded C
- Frameworks and Tools: VS Code, Git, PySpark, Boto3,
- Terraform, AWS Serverless Services, Data Lake
- Certifications - AWS Data Engineer, AWS Cloud Practitioner

Achievements

- Pioneered Cost-Saving Initiatives
- Leadership in Project Delivery
- Process Automation for Time-Savings
- Streamlined Deployment Pipelines minimizing manual intervention

Projects

Enterprise Medallion Data Lake Platform

- Architected a production-grade Medallion Architecture with three layers: a Raw landing zone using Apache Iceberg on Amazon S3 Tables for ACID transactions.
- an Exploratory layer via Amazon Athena and AWS Glue Data CatLog and a Curated layer in Amazon Redshift for business-ready aggregations.
- Built a governance layer over the data lake using AWS Lake Formation for tag-based column/row-level access control and cross-account permissions

End-to-End Serverless Architecture for Data Integration

- Designed and implemented an end-to-end serverless architecture using AWS Lambda, API Gateway, SQS, DynamoDB, etc. for seamless data integration across multiple customers.
- Developed APIs to enable real-time data consumption and production to downstream systems.
- Schema evolution with zero pipeline downtime via Iceberg schema registry

AI Chatbot

- Developed an AI-powered chatbot using Python, RAG architecture, and Groq-hosted LLMs to deliver context-aware and accurate responses from custom knowledge sources.
- Implemented document ingestion, chunking, and vector-based semantic retrieval to enhance response relevance and reduce hallucinations in chatbot interactions.
- Integrated Groq LLM APIs with a Retrieval-Augmented Generation pipeline for low-latency inference and efficient question-answering workflows.
- Built an interactive user interface using Streamlit, enabling real-time chat, document upload, and conversational query handling.
- Optimized prompt engineering and retrieval logic to improve answer precision, response speed, and overall user experience.